MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, RIGHT-ANGLE, HERMAPHRODITIC CONTACT FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connectors described herein shall consist of this document and the latest issue of Specification MIL-C-55302.

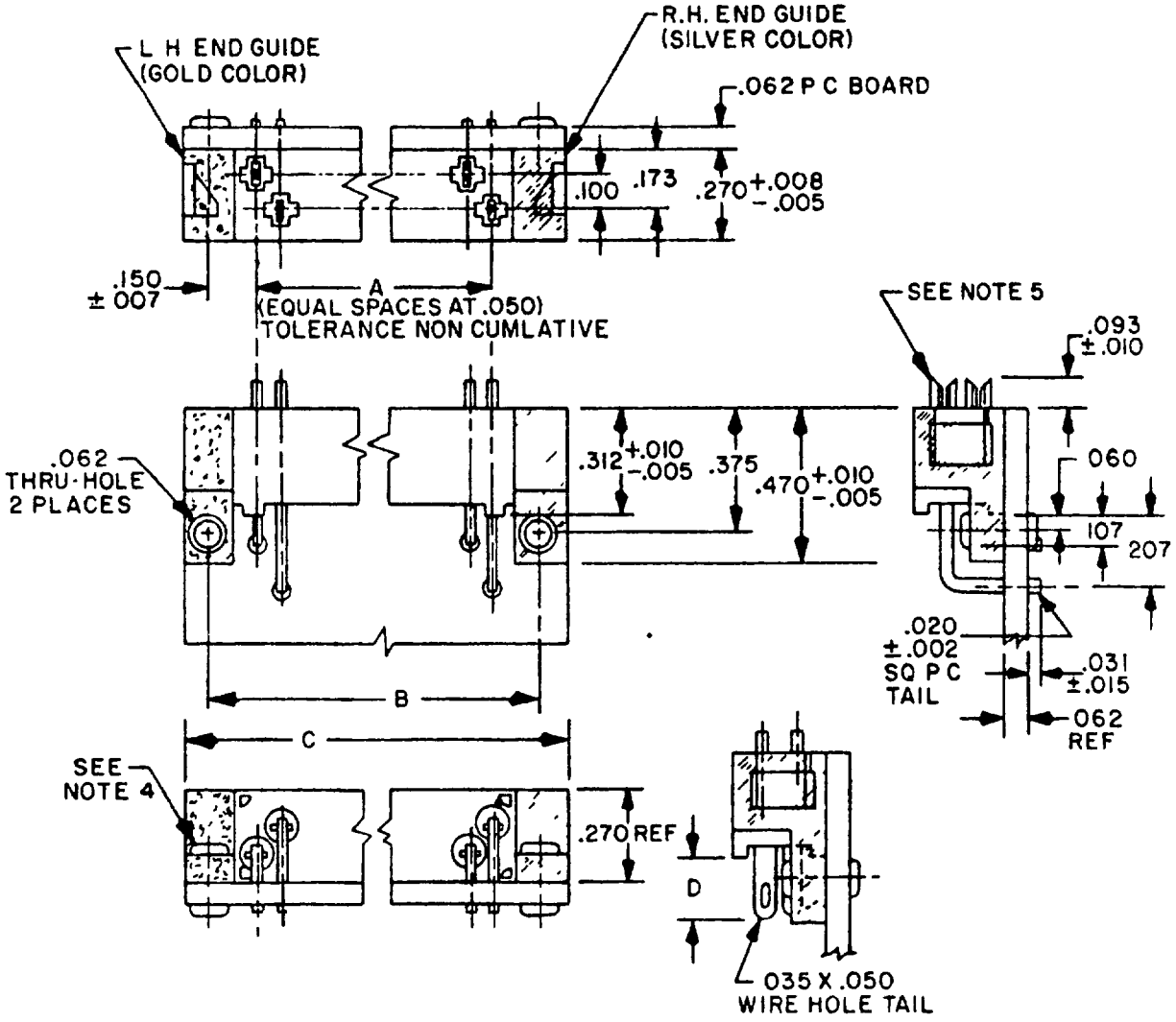
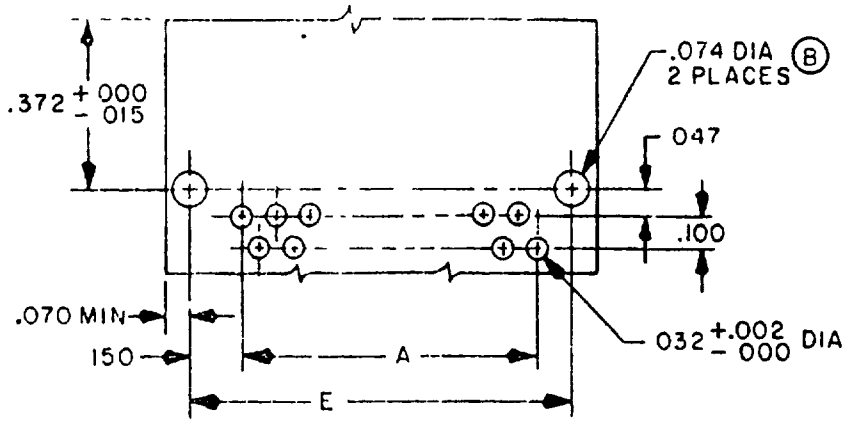


FIGURE 1 Connector, plug (.100 spacing).

(D) denotes changes

FSC 5935

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DETAIL OF P C CARD SHOWING
RECOMMENDED PLUG HOLE PATTERN

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.035	.89	.107	2.72	.470	11.94	2 350	59.69
.005	13	.047	1.19	.150	3.81	.850	21.59	2.490	63.25
.007	.18	.050	1.27	.157	3.99	1.150	29.21	2.650	67.31
.008	.20	.060	1.52	.173	4.39	1.290	32.77	2.950	74.93
.010	.25	.062	1.57	.207	5.26	1.450	36.83	3.090	78.49
.015	.38	.070	1.78	.270	6.86	1.750	44 45	3.550	90.17
.020	.51	.074	1.88	.312	7.92	1.890	48 01	3.850	97.79
.031	.79	.093	2.36	.372	9 45	2.050	52.07	3 990	101.35
.032	.81	.100	2.54	375	9.52	2 190	55.63		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25 4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm).
4. Connectors are supplied with open eyelets (2), which are to be rolled over after assembly to printed wiring board.
5. With connectors fully engaged the mating contacts shall have a minimum overlap of .062 inches. As a minimum the contacts shall exhibit a wiping action through this distance

FIGURE 1 Connector, plug (100 spacing) - Continued

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TABLE I Dash number and dimensions. (B)

Dash number 1/	No. of contacts	Dimensions (inches)				
		A	B ±.010	C Ref.	D ±.015	E
01	18	.850	1.150	1.290	.157	1.150
02	18	.850	1.150	1.290	---	1 150
03	30	1.450	1.750	1.890	.157	1.750
04	30	1 450	1.750	1.890	---	1.750
05	36	1.750	2.050	2.190	.157	2.050
06	36	1.750	2.050	2.190	---	2.050
07	42	2.050	2.350	2.490	.157	2.350
08	42	2.050	2.350	2.490	---	2.350
09	54	2.650	2.950	3.090	.157	2.950
10	54	2.650	2.950	3.090	---	2.950
11	72	3.550	3.850	3.990	.157	3.850
12	72	3.550	3.850	3.990	---	3.850

1/ See requirements for complete part number when polarization is required.

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REQUIREMENTS:

Dimensions and configuration. See figure 1 and table I.

Material and finish.

(B) Contact 0.20 inch thick phosphor bronze per QQ-B-750, composition "A". Gold plate per MIL-G-45204, type II, grade D, class 1, over nickel plating per QQ-N-290, class 2, 30 to 150 microinches in the engagement area for a length of .120 inch minimum (see figure 1). The remainder of the contact shall be gold plated per MIL-G-45204, type II, grade D, 10 microinches minimum over nickel plating per QQ-N-290, class 2, 30 to 150 microinches.

Insulator. MIL-M-14, type SDG-F.

End guides: Left hand, sintered brass 90/10, clear chromate per MIL-C-5541, class 1A (gold color). Right hand, sintered brass 90/10, nickel plate 300 microinches per MIL-F-14072 (silver color).

Polarizing pin: Aluminum alloy 2011-T3 with clear chromate, per MIL-C-5541, class 1A.

Contact identification Numerical sequence starting with the closest contact adjacent to the left hand end guide (gold color) shall be contact number 1, contact number 2 would be the next offset contact, etc.

Mating and unmating: The maximum insertion force, in pounds, shall not exceed a value equal to .5 times the number of contacts, and the withdrawal force, in pounds, shall be a minimum of .11 times the number of contacts and shall not exceed the measured insertion force.

(B) Individual contact separation force:

Unplated contacts: One ounce minimum when tested in production with gage shown on MIL-C-55302/97.
Plated contacts: One ounce minimum.

Contact resistance: The contact resistance of all pairs measured shall not exceed 20 milliohms.

Low level circuit. The low level circuit resistance shall not exceed 20 milliohms.

Dielectric withstanding voltage.

Sea level. 1,000 volts rms, 60 cycles, ac.
High altitude (70,000 feet): 300 volts rms, 60 cycles, ac.

Current rating, maximum: 5 amperes.

Polarizing pin: When required, one of the following codes shall be added to the dash number:

"P" - specifying location by contact number where polarizing pin shall be inserted; for example, M55302/53-01P17 (polarizing pin inserted in place of contact number 17).
"H" - specifying location by contact number where contact shall be omitted for mating, for example, M55302/53-01H17 (polarizing hole has contact number 17 omitted).

(B) Mating receptacle: Shall conform to MIL-C-55302/52, /99, and /102.

Part number: M55302/53- (dash number from table I).

Patent number 2,750,572. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

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Custodians
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS
Air Force - 11, 99
DLA - ES

User activities
Army - AT, AV
Navy - MC, OS, SH
Air Force - 19

Preparing activity
Army - CR

Agent
DLA - ES

(Project 5935-3145-2)

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DOCUMENT IDENTIFIER (Number) AND TITLE
MIL-C-55302/53B Connectors, Printed Circuit Subassembly and Accessories: Plug Right Angle, Hermaphroditic Contact for Printed Wiring Boards (.100 Spacing)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

HQ, USA CORADCOM
ATTN: DRDCO-CM-DM
Ft. Monmouth, N.J., 07703

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